Cross-Border Return Transmission from the US Session to European and Japanese Equity Markets

SIQuant Research[1]

[1]Hypothesis × Quant

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Abstract

Hypothesis: $H_0 : \beta_{\text{US}\to\text{Asia/Europe}} = 0$ versus $H_1 : \beta > 0$ for the next-session response of the Nikkei 225 and Euro Stoxx 50 to the prior S&P 500 return
Data: Daily index returns, 2016-10-25 to 2026-10-23, 2,304 sessions common to all three markets
Method: OLS with Newey–West HAC errors (5 lags), a 2000-draw circular-shift placebo, and a Cholesky-FEVD spillover index from a BIC-selected VAR(1)
Results: Nikkei: $\beta = 0.500$, $t = 7.70$, placebo $p = 0.0005$; Euro Stoxx: $\beta = 0.118$, $t = 1.93$, placebo $p = 0.0290$. Decision threshold: Bonferroni $\alpha = 0.025$.

1 Hypothesis and Design

Asian and European sessions close before the US session, so the previous US return is strictly predetermined [Engle et al., 1990]. The test is a one-sided HAC regression with a circular-shift placebo that preserves serial dependence while breaking the cross-market alignment.

2 Results

The FEVD total spillover index is 22.4% with ordering US, EU, JP and 24.5% with the reverse ordering; Cholesky FEVD is order dependent, so the gap between the two is reported as sensitivity [Diebold and Yilmaz, 2012].

Table 1: Next-session response to the prior US session (HAC Newey–West, one-sided $H_1 : \beta > 0$)

Target	β	HAC t	p	Placebo p	R^2	N
Nikkei 225 (t)	0.5004	7.70	6.76e-15	0.0005	0.1799	2,303
Euro Stoxx 50 (t)	0.1182	1.93	2.66e-02	0.0290	0.0142	2,303

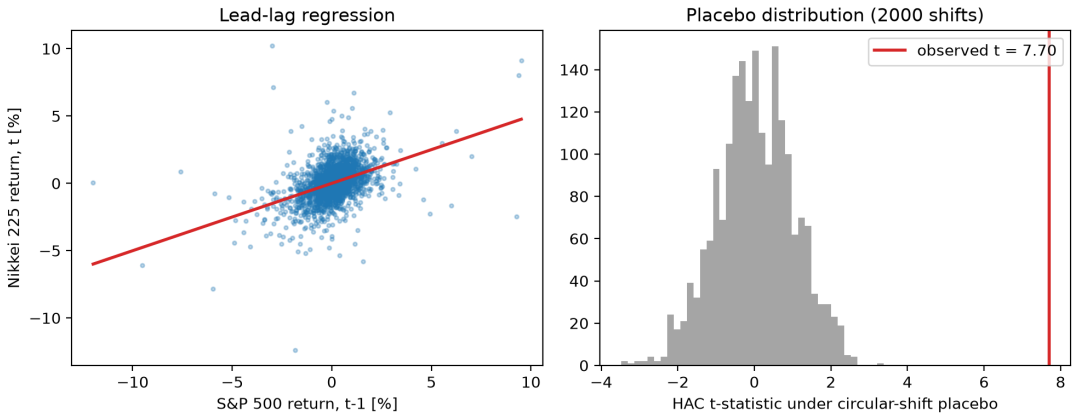


Figure 1: Lead-lag fit and placebo distribution.

3 Limitations

Only sessions present in all three calendars are used; no transaction costs or trading rule are evaluated; returns rather than realised volatility are modelled.

References

Francis X Diebold and Kamil Yilmaz. Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1):57–66, 2012.

Robert F Engle, Takatoshi Ito, and Wen-Ling Lin. Meteor showers or heat waves? heteroskedastic intra-daily volatility in the foreign exchange market. *Econometrica*, pages 525–542, 1990.